



Closed-loop enhancement and neural decoding of cognitive control in humans

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Deficits in cognitive control—that is, in the ability to withhold a default pre-potent response in favour of a more adaptive choice—are common in depression, anxiety, addiction and other mental disorders. Here we report proof-of-concept evidence that, in participants undergoing intracranial epilepsy monitoring, closed-loop direct stimulation of the internal capsule or striatum, especially the dorsal sites, enhances the participants' cognitive control during a conflict task. We also show that closed-loop stimulation upon the detection of lapses in cognitive control produced larger behavioural changes than open-loop stimulation, and that task performance for single trials can be directly decoded from the activity of a small number of electrodes via neural features that are compatible with existing closed-loop brain implants. Closed-loop enhancement of cognitive control might remediate underlying cognitive deficits and aid the treatment of severe mental disorders.

Mental disorders are a leading source of medical economic burden¹. Current therapies do not target the cause of these disorders: dysfunctional communication in distributed brain circuits^{2,3}. Electrical brain stimulation can effectively modulate such circuits⁴, but clinical neurostimulation trials in mental disorders have had mixed results^{5,6} for two reasons. First, spatially targeting a circuit may not be enough. Symptom-related activity is only occasionally present, and effective therapies may require temporal specificity: a closed feedback loop that detects pathological activity and intervenes to restore more efficient function^{7–9}. Second, mental illness is heterogeneous. A label such as 'depression' likely describes several different brain disorders^{8,10}. Rather than trying to detect or treat ill-specified symptoms such as mood^{11,12}, it may be more effective to target objective, rigorously measurable constructs, each representing a different component of cognitive or emotional processing^{8,10,13}. In this dimensional approach, simultaneous impairments in multiple cognitive constructs act as 'ingredients' to produce the clinical manifestation of disease^{2,10}. An advantage of the dimensional approach is that individual constructs can be objectively measured, for example, through performance on standardized tasks. That performance may be more easily reducible to specific brain circuits and stimulation targets.

One particularly promising construct is cognitive control, the ability to flexibly alter strategies or response styles as goals change^{14–18}. First, cognitive control deficits are common across mental disorders, including depression, addiction, schizophrenia and obsessive-compulsive disorder¹⁹. They are particularly relevant for mood and anxiety disorders, where patients are often unable to disengage from habitual, distress-driven behaviour. Second, key aspects of control are readily measurable, using cognitive conflict

tasks. Slower response times in these tasks signify difficulty exerting control to overcome response conflict, to the point that performance on such tasks is the most commonly used clinical and laboratory metric of cognitive control^{19,20}. Third, control and conflict have a likely circuit substrate. Action selection in the face of competing options is well linked to interactions between multiple subregions of the prefrontal cortex (PFC) and their counterparts in the striatum^{21–23}. They similarly have a physiologic marker: many different conflict tasks evoke robust electrophysiologic signatures, namely theta (4–8 Hz) oscillations in PFC^{20,24,25} and low-frequency synchrony among frontal and subcortical structures^{19,26–28}. Fourth, that circuit substrate is well suited for manipulation through brain stimulation. The PFC, striatum and basal ganglia are densely interconnected by white matter tracts that run largely in the internal capsule^{29,30}. Those tracts follow relatively stereotyped trajectories^{29–32}, meaning that stimulation at a given capsular site will likely engage similar circuitry across individuals. To that point, deep brain stimulation (DBS) of the internal capsule and of the subthalamic nucleus, as well as transcranial current stimulation of the lateral PFC all improve performance on cognitive control tasks^{18,33,34}. In other words, the scientific basis exists for direct remediation of cognitive control deficits, if these disparate findings can be integrated.

Three barriers have prevented that integration. First, there are no extant techniques for quantifying cognitive control performance in real time. Response time on conflict tasks, the standard metric, is also influenced by stochastic noise and often by changes in conflict level between subsequent task trials³⁵. Second, there are no proven ways to rapidly alter that performance, that is, to respond to and cancel out momentary lapses. Third, although there are known physiologic signatures of cognitive control when averaged

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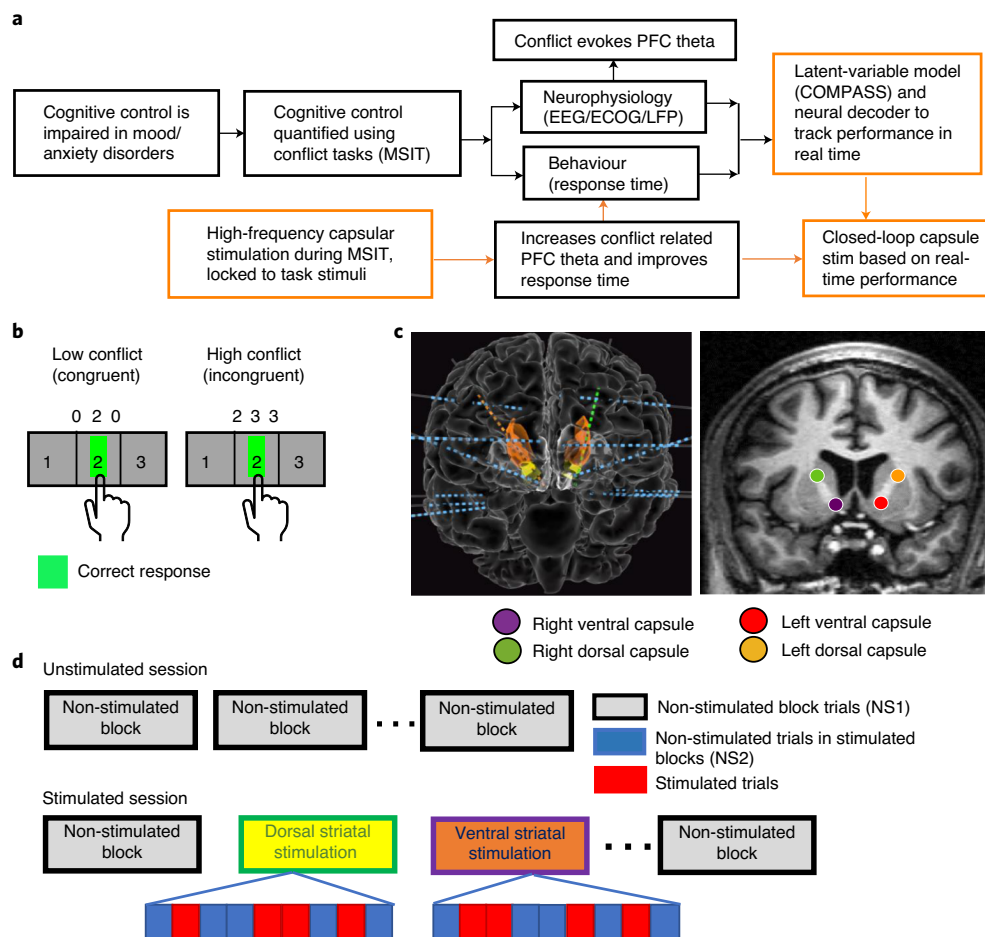


Fig. 1 | Experimental paradigms. a, Overview of cognitive conflict/control processing and the current study. Conflict, which requires high control, evokes PFC theta oscillations and slows decision (response) times. Orange boxes highlight this paper's work. We developed intermittent stimulation approaches that improve these physiologic and behavioural correlates of control. We also developed a state-space model and neural decoder to track these quantities in real time. We then merged these for closed-loop enhancement of cognitive control, on the basis of state-space behaviour tracking. **b**, Schematic of the MSIT, in which participants must inhibit pre-potent responses on 50% of trials. **c**, Schematic of a typical montage of depth electrodes (left) and stimulation targets (right); the caudate nucleus and nucleus accumbens in **c** (left) are coloured orange and yellow, respectively. **d**, Trial structure during experimental sessions. In an unstimulated experimental session (top), blocks of trials were separated by brief rest periods. During an open-loop stimulation experiment (bottom), some blocks of MSIT trials had no stimulation at all (NS1), while others had stimulation only on a randomly selected 50% of trials. Non-stimulated trials in these blocks are designated NS2. Dashed lines indicate SEEG electrode trajectories. ECOG, electrocorticography.

across time, there are no such signatures for moment-to-moment fluctuation in control. Here we demonstrate methods to overcome those barriers, through a proof of concept in participants undergoing stereotaxic electrode monitoring for epilepsy (Fig. 1a, Extended Data Fig. 1). We developed a method for quantifying conflict task performance at the single-trial level, while regressing out sources of undesired variability. This method is grounded in a state-space or latent-variable formalism, which generalizes the Kalman filtering approach that has been successful in other neural interface applications^{36,37}. We showed that brief internal capsule stimulation rapidly enhances cognitive control. That enhancement was visible in raw task performance, in our latent variables and in PFC theta oscillations. We then integrated those findings into a closed-loop controller that was stimulated in response to lapses in cognitive control. Closed-loop stimulation produced larger performance changes than a corresponding open-loop paradigm. Finally, we showed that task performance, at the scale of single trials, can be decoded entirely from brain activity. Together, these findings provide proof of concept for a closed-loop system for treating cognitive control deficits.

Results

Cognitive control and theta power enhancement through internal capsule stimulation. Participants performed a cognitive control task (the Multi-Source Interference Task (MSIT), Fig. 1b) while undergoing invasive multi-electrode monitoring. During task performance, we electrically stimulated in the internal capsule at similar sites across participants (Fig. 1c). We collected 8,790 trials across 176 blocks from 21 participants – 12 without brain stimulation, 5 with both non-stimulated and open-loop stimulation sessions, 1 with only open-loop stimulation and 3 with non-stimulated and closed-loop stimulation sessions. Dropping incorrect responses and non-responses excluded 345 trials (3.92% of total; 8,445 trials retained in analysis). In open-loop experiments, a random 50% of trials received brief, task-linked stimulation (Fig. 1d).

MSIT engaged cognitive control: participants were 216 ms slower on high-conflict trials ($N=21$, $P<0.001$, $t=33.62$, Wald test on model coefficient; Fig. 2a). Conflict increased task-related theta power in the posterior cingulate ($P<0.02$, $t=3.24$) and in the dorsolateral prefrontal cortex ($P<0.001$, $t=4.31$) (Fig. 2b). Capsular stimulation enhanced both cognitive control and its elec-

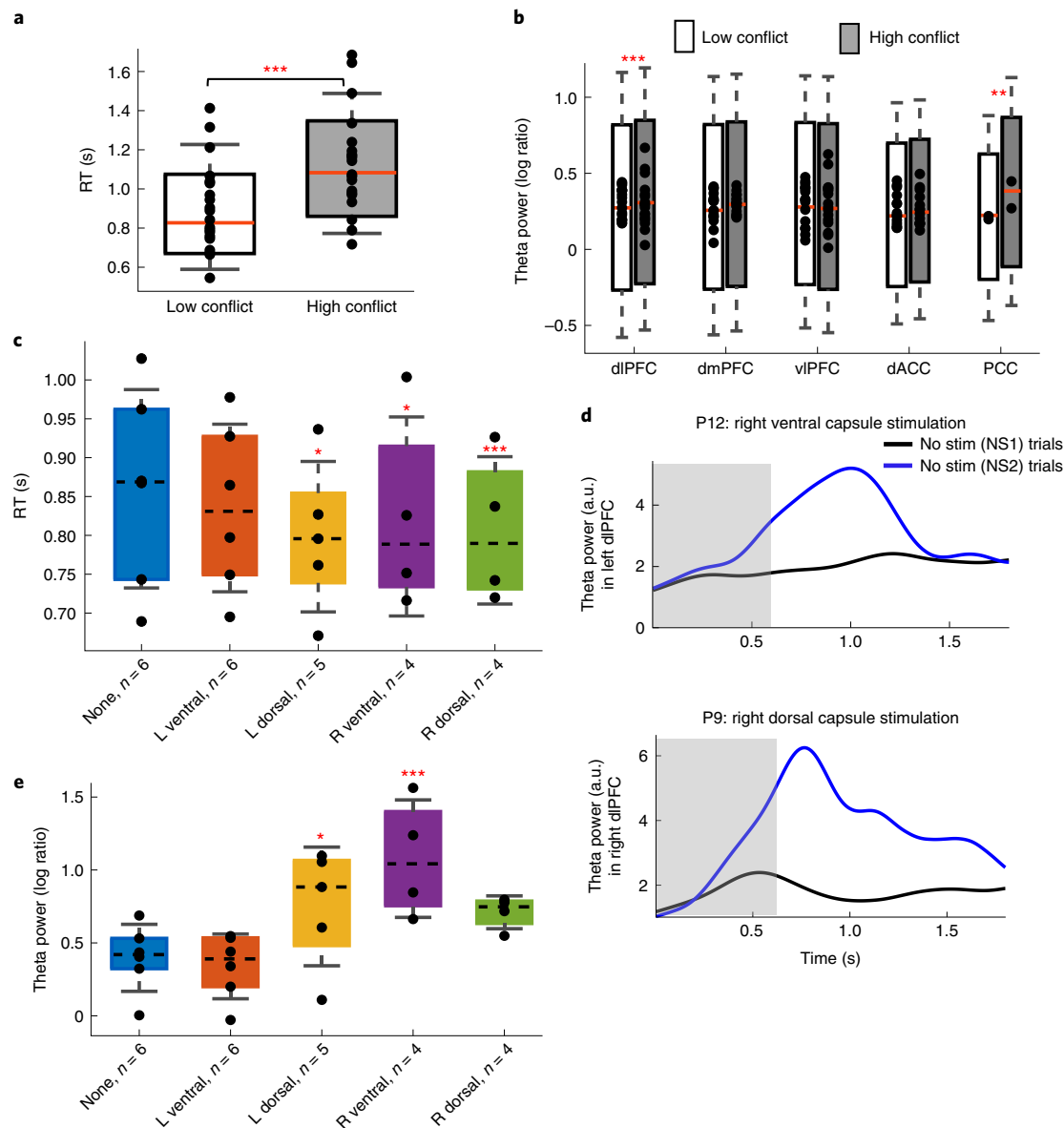


Fig. 2 | Effect of conflict and open-loop capsular stimulation on cognitive control. **a**, RTs during low-conflict and high-conflict non-stimulated trials (NS1) from 21 participants show significant differences ($P < 0.001 = 7.84 \times 10^{-124}$, calculated at the single-trial level across 21 participants). Data points show the mean for each condition for individual participants. **b**, Theta power during low-conflict and high-conflict NS1 trials in frontal regions. Power is expressed as the log ratio relative to pre-trial background. As in earlier studies, conflict slowed responses and evoked significantly higher theta in dlPFC ($P < 0.001$) and PCC ($P = 0.006$), P values calculated at the single trial level. Data points show the mean for each condition for individual participants. **c**, RT during open-loop stimulation during MSIT. In **a–c**, the dots represent the individual participants, the bars show the median (red line or black dashed line) and the bar maxima/minima correspond to the 75th and 25th percentiles, respectively; error bars show s.e.m. Stimulation improved task performance, as reflected in significantly lower RTs with L Dorsal ($P = 0.015$), R Ventral ($P = 0.031$) and R Dorsal ($P < 0.001$) capsular stimulation. Colours correspond to stimulation sites in Fig. 1c, and inferential testing was performed against the non-stimulated condition. **d**, Example theta power traces from PFC channels in two participants (P9 and P12), with right ventral (top) and right dorsal (bottom) capsular stimulation. NS1 and NS2 trials are as in Fig. 1d. The stimulation was from 0.90.6 s (grey window), although note that the curves show only data from non-stimulated trials. **e**, Task-evoked theta power (log ratio versus pre-trial baseline) across PFC channels in NS1 trials (None) compared to NS2 trials (all other conditions). L Dorsal and R Ventral capsular stimulation significantly increased theta power ($P = 0.043$, $P < 0.001$, respectively), as in earlier reports¹⁸. Number of subjects (n) for each stimulation type is specified on the X axis. In all panels, P values (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$) are reported after correcting for multiple comparisons using an FDR. dlPFC, dorsolateral PFC; dmPFC, dorsomedial PFC; vlPFC, ventrolateral PFC; dACC, dorsal anterior cingulate cortex; PCC, posterior cingulate cortex.

trophysiologic correlates, with dorsal sites showing stronger effects. Right dorsal ($P < 0.001$, $t = -4.28$), left dorsal ($P < 0.01$, $t = -2.65$) and right ventral ($P < 0.05$, $t = -2.64$) capsular stimulation all significantly decreased reaction time (RT, Fig. 2c and Supplementary Table 3a) without impairing accuracy (Extended Data Fig. 2a). RTs under dorsal stimulation were faster than with ventral stimulation

on both sides, with right dorsal stimulation being the overall most effective (Supplementary Table 3a). These findings mirror the capsular topography, where more dorsal sites are enriched in fibres originating in the dorsal PFC^{29,30}, which in turn is associated with cognitive control during conflict tasks^{19,20,27,38}. Consistent with that topography, stimulation in dorsal capsule sites outside of task per-

formance propagated mainly to the anterior cingulate cortex and the dorsolateral PFC (dlPFC, Extended Data Fig. 3). RT improvements were not explained by practice or regression to the mean; RT in the final non-stimulated (NS1) block was significantly higher than that in the initial block (Supplementary Fig. 1a).

There was no evidence for an interaction between stimulation and conflict level (Akaike's Information Criterion (AIC): -449.27 for a model without an interaction term versus -445.72 with interaction). To assess stimulation effect on theta, we analysed artifact-free trials interleaved within stimulated blocks (NS2, Fig. 1d) and compared these to blocks without stimulation (NS1). Left dorsal and right ventral capsular stimulation significantly increased theta power in NS2 compared with NS1 trials (L Dorsal: $P=0.0428$, R Ventral: $P=0.0006$, false discovery rate (FDR)-corrected; Fig. 2d,e and Supplementary Table 3b). Right dorsal capsular stimulation also increased theta but did not reach significance ($P=0.1733$, FDR-corrected). Theta increases were present in many PFC channels and were specific to behaviourally effective stimulation (Supplementary Fig. 2). Theta increases also could not be explained as regression to the mean or practice effects, as theta power attenuated over the experimental session in the absence of stimulation (Supplementary Fig. 1B).

Closed-loop stimulation based on a state-space model efficiently enhances cognitive control. We next sought to quantify task performance and capsular stimulation effects at a trial-to-trial level. We achieved this with a state-space latent-variable model (Fig. 3a,b). This model assumes that the RT of each trial can be modelled as the combination of a baseline or expected RT for all trial types (x_{base}), a specific slowing on high-conflict trials (x_{conflict}) and a log-Gaussian noise process. This formulation allows rapid tracking of stimulation-induced changes (see Methods). We verified that the two-process model captured the majority of variance in the RT data (Supplementary Fig. 3) and converged in all participants (Supplementary Fig. 4), and that response accuracy did not carry additional information (Supplementary Fig. 5). Stimulation in the dorsal capsule improved overall performance (x_{base} , Fig. 3c) and reduced conflict effects (Fig. 3d). Right dorsal capsular stimulation again had the largest effects. Ventral stimulation significantly reduced x_{conflict} , but not x_{base} . The observed differences could not be explained by block-to-block carry-over or other persistent effects, and x_{base} rapidly increased once stimulation ceased (Supplementary Fig. 6).

We applied capsular stimulation under closed-loop control in three further participants. We estimated x_{base} in real time and triggered stimulation during control lapses, that is, when x_{base} increased beyond an experimenter-determined threshold (Fig. 4a). As predicted, conditioning stimulation on x_{base} specifically improved that variable (Fig. 4b) without enhancing x_{conflict} (Fig. 4c). Closed-loop stimulation was more effective than open-loop stimulation. Stimulation of the right ventral capsule, which did not have significant effects in open-loop tests (Fig. 3c), now significantly reduced x_{base} ($P<0.01$, permutation test; Fig. 4b). At both dorsal stimulation sites, closed-loop stimulation reduced x_{base} significantly more than open-loop stimulation ($P<0.001$, permutation test; Fig. 4b). There were again no accuracy effects (Extended Data Fig. 2b). Closed-loop stimulation effect was manifested in raw RT data for right dorsal capsule stimulation ($P<0.001$, permutation test; Extended Data Fig. 4). The effects cannot be explained by regression to the mean (Supplementary Fig. 7). The closed-loop algorithm stimulated more often than its open-loop counterpart (22–29 stimulated trials per block), but this alone did not explain the increased behaviour effect. Closed-loop stimulation also appeared more efficient than open-loop stimulation in terms of performance gain for the applied energy. It produced a greater change in x_{base} per stimulation in the right ventral and dorsal capsules (Fig. 4d), although this

did not reach the pre-determined significance threshold (R Ventral: $P=0.207$, R Dorsal: $P=0.293$).

A few participants reported that improving objective task performance also improved their subjective well-being. Although participants could not directly identify when stimulation was on or off, they perceived when they were performing more fluidly (Supplementary Table 4). Two participants who had previously reported difficulty with effortful self-control noted relief of their usual anxiety (Supplementary Table 5). No participant reported negative emotional effects during any stimulation experiment.

Neural decoding of cognitive states for closed-loop control. To demonstrate that cognitive control lapses could be remediated outside of a controlled, structured task setting, we developed decoders to read out cognitive control from local field potentials (LFP). For each participant, we estimated an encoding model (Fig. 5a) to map cognitive states to LFP power. State variables were linearly related to neural features (Supplementary Fig. 8) at a level exceeding chance (Supplementary Fig. 9). The confidence intervals of cognitive states decoded from LFP and estimated from behaviour largely overlapped (Fig. 5b; x_{base} : $84.02 \pm 15.8\%$ overlap, x_{conflict} : $83.17 \pm 16.3\%$ overlap). Decoding used relatively few power features in each participant (Fig. 5c; 11.75 ± 6.63 features for x_{base} and 11.27 ± 6.74 for x_{conflict}). Decoding weighted brain regions commonly implicated in cognitive control. x_{base} was encoded primarily in dlPFC (8–30, 130–200 Hz), vlPFC (15–30, 65–110 Hz) and temporal cortex (multiple bands). x_{conflict} was similarly encoded in dlPFC (most bands), vlPFC (8–55 Hz), temporal cortex (broadband) and amygdala (8–55 Hz).

Decoding was also possible during intermittent brain stimulation (x_{base} : $83.15 \pm 9.04\%$, x_{conflict} : $84.43 \pm 9.6\%$ of trials overlapping the confidence interval of the behavioural estimate). Stimulation did not meaningfully alter the number of features needed for decoding (x_{base} : 11.11 ± 4.73 versus 11.75 ± 6.63 features; x_{conflict} : 12.55 ± 4.24 versus 11.27 ± 6.74 features; all $P>0.5$, unpaired t -test). However, it decreased the number of cortical regions that encoded either x_{base} or x_{conflict} (Fig. 5d). This encoding may have transferred to the dorsal striatum (caudate), which showed increased encoding across frequency bands, although this did not reach our pre-specified significance threshold for x_{base} (x_{base} : $t=-0.3980$, $P=0.6908$, x_{conflict} : $t=5.3947$, $P<0.001$, paired t -test between encoder model coefficients on NS1 and NS2 trials across participants).

Discussion

Cognitive control is impaired in numerous mental disorders^{8,13,19,39}. We augmented one aspect of human cognitive control, performance on a cognitive conflict task, by intermittent closed-loop stimulation of the internal capsule. The effects were detectable in both manifest data (raw reaction times) and derived variables. This brief intervention produced positive emotional effects in a small subset of patients who reported relevant pre-existing impairments. This may suggest a future utility of cognitive control as a psychiatric treatment target, independent of a patient's diagnostic label. To that point, we previously showed that improvements in similar aspects of cognitive control may be a mechanism of action of clinical deep brain stimulation¹⁸. Further, we could separate and alter components of cognitive control. We enhanced baseline performance (expected RT on all trial types) without driving the conflict response (specific slowing and expected RT on high-conflict trials) in the same direction. This suggests that these two processes could be targeted separately, for example, if only one was impaired in a hypothetical future patient. Both states could be decoded with 11 LFP spectral features per participant, from a mean of 6 brain regions. Existing human-grade implants may be able to implement these or similar decoders⁴⁰.

Our result both converge with and diverge from past studies of cognitive control. Anatomically, we found the largest effects from

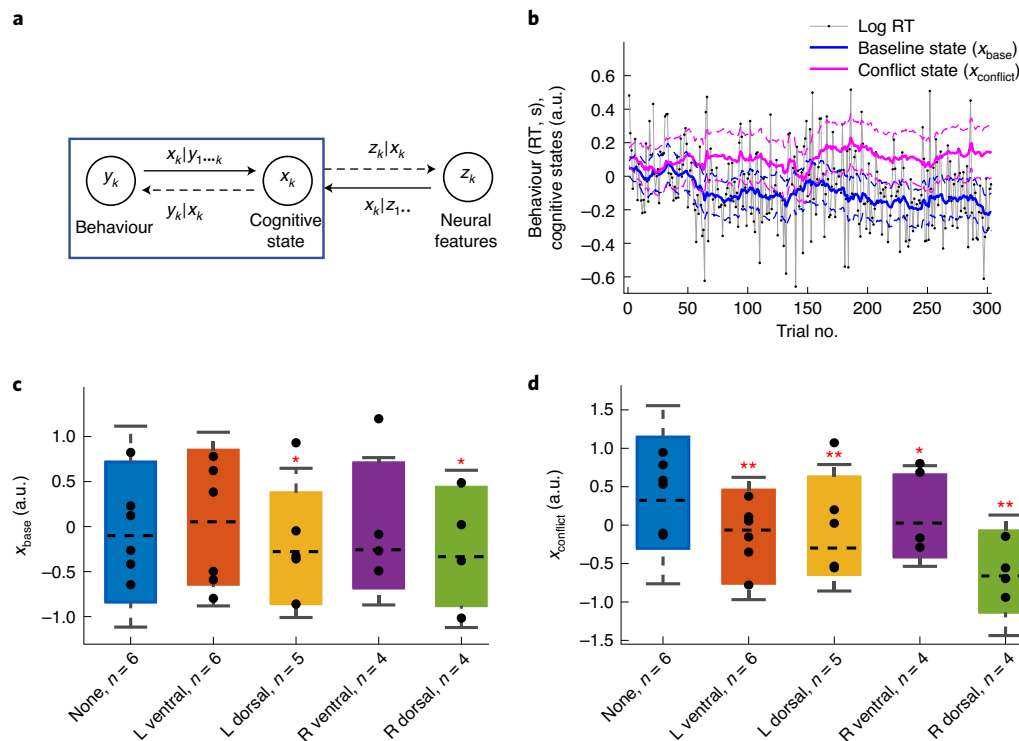


Fig. 3 | Effect of open-loop capsule stimulation on cognitive control. **a**, Schematic of the modelling framework, where behaviour and neurophysiology are linked through a low-dimensional latent state space. Here we focus on inferring latent states from behaviour (blue box). **b**, Example of a participant's raw behaviour (RT) and its decomposition into x_{base} and $x_{conflict}$. **c**, Effect of open-loop, randomly interleaved stimulation on x_{base} (expected RT). Only dorsal stimulation significantly improved this task component ($P = 0.026$). **d**, Effect of the same stimulation on $x_{conflict}$ (expected RT to high conflict trials). Stimulation in all internal capsule sites altered this aspect of cognitive control (R Ventral: $P = 0.02$, rest: $P < 0.001$). Number of subjects (n) for each stimulation type is specified on the X axis. In **c** and **d**, P values ($*P < 0.05$, $**P < 0.001$) are reported after correcting for multiple comparisons using an FDR. Statistical inference was done through non-parametric permutations due to the highly autocorrelated state variables (see Methods). Data points represent individual participants, bars show the median (dashed line) and the bar maxima/minima correspond to the 75th and 25th percentiles, respectively; error bars show s.e.m.

stimulation in the dorsal capsule. This part of the capsule contains corticofugal axons originating mainly in the lateral and dorsomedial PFC and the cingulate cortex³⁰. We observed that the lateral PFC in particular showed a theta power enhancement during behaviourally effective stimulation. This fits with the central role that those PFC regions likely play in cognitive control. We obtained our largest effects by stimulating in areas where we could modulate the relevant PFC subregions through retrograde activation. Our physiologic results support that retrograde model – effective stimulation increased PFC theta power, the most commonly reported correlate of control. Surprisingly, however, our decoders did not strongly select theta-band features when given the full range of LFP activity to consider. This may be because prior studies linking cognitive control to theta used electroencephalography (EEG)²⁰ and were analysed on a trial-averaged basis. The mean theta effect across trials, although robust, may not relate to the trial-to-trial performance. Further, LFP signals from a few millimetres of tissue may not correlate well with EEG, which records synchronized signals from broader swaths of cortex. Additionally, state variables were strongly encoded outside the canonical PFC–cingulate cognitive control network^{19,38}, particularly in the temporal lobe. Hippocampus and amygdala are important in human anxiety-like behaviour⁴¹ and integrate signals from multiple prefrontal regions⁴². These structures may play a greater role in cognitive control than has been recognized.

Effective stimulation shifted encoding from the frontotemporal cortex to the dorsal striatum or caudate. This is consistent with models of cortico-striatal function that consider the dorsomedial

striatum, including the caudate, to be part of a loop-structured circuit serving flexibility and exploration^{19,21}. That encoding may also explain why dorsal capsular stimulation was more behaviourally effective, as the stimulation field likely overlapped both caudate and incoming fibres from dorsal PFC. It is less clear why stimulation slightly decreased the overall encoding of control-related variables, yet improved task performance. One possibility is the hypothesized link between cognitive control and anxiety⁴³. Humans may normally exert more cognitive control than we need to, that is, over-weighting risk. Our performance enhancement could be explained as a reduction of that ‘excess’ control. This is concordant with the two participants who reported that stimulation allowed them to shift attention away from anxious, self-focused processing. We note that this is also a very different subjective effect than reports from clinical DBS of the capsule or striatum, which usually causes euphoria^{44,45}. It seems unlikely that such positive mood changes drove faster RTs because the majority of participants reported no subjective mood alteration.

Because participants had seizure vulnerability, we were limited to a single stimulation session between the completion of clinical monitoring and electrode removal. We could not retest the same stimulation multiple times in the same participant, nor could we attempt a variety of setpoints to determine the limits of this closed-loop approach. We were limited to intermittent stimulation to reduce seizure risk and could not directly test this intermittent paradigm against the continuous DBS of previous studies. Electrodes were placed along individualized trajectories, and their location within the capsule varied substantially (Supplementary Fig.

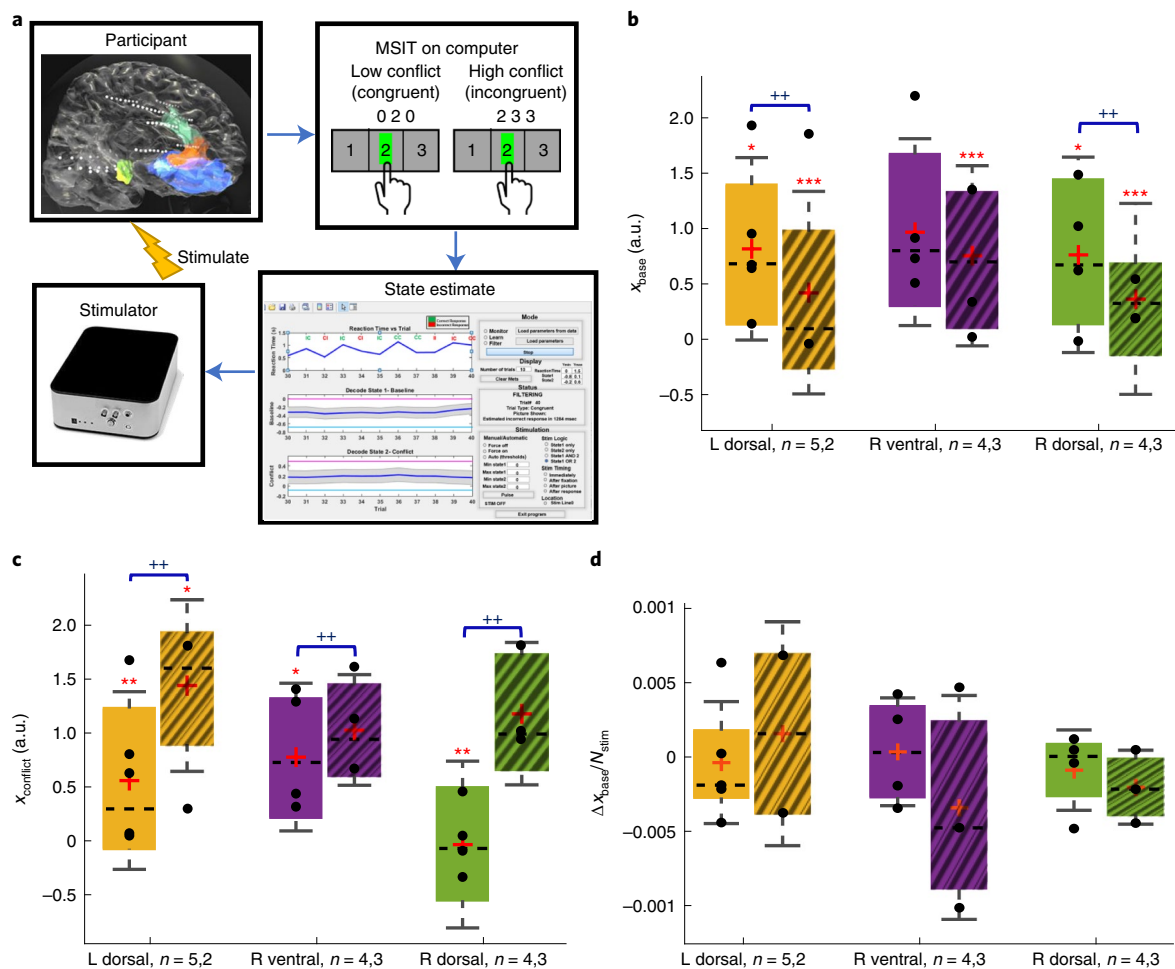


Fig. 4 | Closed-loop internal capsule stimulation efficiently enhances cognitive control. **a**, Schematic of closed-loop paradigm. Briefly, the participant with 100+ electrodes recording LFP (top left) performed MSIT (top right) on a desktop computer. A real-time state estimator based in MATLAB (bottom right) calculated x_{base} and $x_{conflict}$ after each trial. This estimator included a threshold-based controller that then triggered stimulation from a neurostimulator (bottom left) on the next trial if the state was above a pre-determined threshold. The real-time state estimates were displayed on a MATLAB-based GUI. **b**, Effect of open-loop stimulation (solid bars) versus closed-loop stimulation (dashed bars) on x_{base} . At dorsal sites within the capsule, closed-loop stimulation was more effective at reducing x_{base} (improving task performance, $***P < 0.001$). **c**, Effect of open- versus closed-loop stimulation on $x_{conflict}$. Stimulation conditioned on x_{base} did not reduce $x_{conflict}$, and in fact significantly increased it at multiple sites ($***P < 0.001$). **d**, Comparison of open- versus closed-loop effects on x_{base} (Δx_{base}) (with 0 representing no change from NS1) divided by the number of stimulated trials (N_{stim}). A negative value indicates a decrease (desired) in x_{base} caused by a specific stimulation on a block level. State values in **b** and **c** are normalized so that non-stimulated blocks have a mean state value of 1 for each participant for both experiments, permitting comparison across participants. Significance was determined by a permutation test, given the highly autocorrelated data. P values (versus no stimulation: $*P < 0.05$, $**P < 0.01$, $***P < 0.001$; versus open-loop stimulation: $++P < 0.001$) are reported after correcting for multiple comparisons using an FDR. Number of subjects (n) for each experiment is specified on the X axis for open- and closed-loop participants. Data points represent individual participants, bars show the median (dashed line) and the bar maxima/minima correspond to the 75th and 25th percentiles, respectively; error bars show s.e.m. Red crosses represent the mean values.

10). In this context, the consistent effect across participants is probably explained by the relatively consistent topographic structure of the capsule across individuals and species^{29,30}. Electrodes used only for recording had a similar limitation. Demonstration of a consistent, repeatable within-subject effect would be critical for turning these results into a therapy. The nearly identical effects between this study and our earlier work with psychiatric DBS patients¹⁸ suggest that the effect is robust. Further, because there is known variability of fibre placement within the capsule despite the topographic map³¹, we might expect much larger effects in a clinical scenario where we could precisely optimize each electrode's placement to engage a target of interest.

We enhanced x_{base} , which reflects overall attentional focus. $x_{conflict}$ corresponds to the more immediate effect of conflict and the dif-

ficulty of executing control. In a clinical setting, either might be disrupted, and we may need to apply closed-loop control to both simultaneously. Here, when we controlled x_{base} , $x_{conflict}$ significantly increased. These two states are not inherently anti-correlated because both were reduced by open-loop stimulation (contrast Fig. 3c,d against Fig. 4b,c), but it remains to be shown that x_{base} can be altered without effects on other cognitive variables. Similarly, our two-state model follows one specific theory, where control is allocated reactively in response to decision conflict. There are other theoretical models of cognitive control measurement and allocation³⁸, and it would be important to explore how our stimulation approaches do or do not alter variables in those frameworks. For instance, alternative frameworks place a heavy emphasis on a participant's motivation to exert control, sometimes framed as expected value. Our

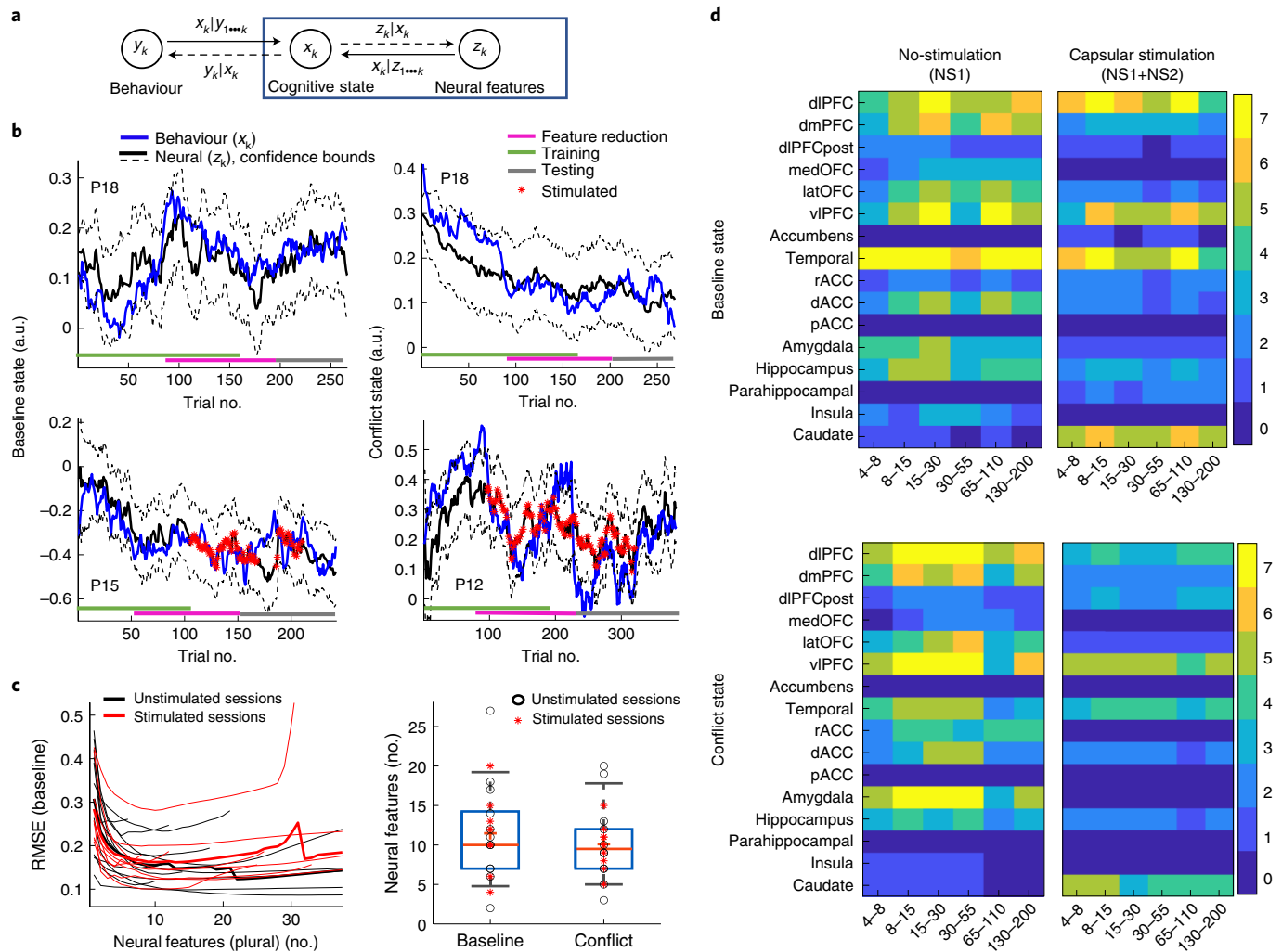


Fig. 5 | Neural decoding of cognitive states. **a**, Schematic of the encoding-decoding framework, which uses the same state variables as Fig. 3 (equation 2). Here we identify linear dependencies between neural features (LFP power) and the latent cognitive states (blue box). **b**, Examples of baseline (left) and conflict (right) states as estimated from behaviour and from neural decoding in subjects P18, P15 and P12. Coloured bars at the bottom of the plot indicate the data segments used for training, feature pruning (validation) and testing. There is strong agreement throughout the task run, including on data not used to train the decoding model. Decoding quality is similar during unstimulated (top) and stimulated (bottom) experiments. **c**, Optimum neural feature set determined using a feature dropping analysis (left). Thin lines represent individual participants ($n=21$), thick lines represent the mean. The open circles indicate the number of features that minimizes root mean square error (RMSE) for x_{base} for each participant on a held-out validation set. This number of features did not differ between x_{base} and x_{conflict} , or between stimulated and unstimulated blocks (right). Data points represent individual participants, bars show the median (dashed line) and the bar maxima/minima correspond to the 75th and 25th percentiles, respectively; error bars show s.e.m. **d**, Number of participants encoding x_{base} (top) and x_{conflict} (bottom) in different brain regions and frequency bands during non-stimulated (left, NS1) and intermittent capsular stimulation (right, NS1+NS2) blocks. Capsular stimulation shifts encoding from cortical regions to subcortical, particularly caudate regions.

results could be explained by an increase in that expected value, and dissecting such questions will require more sophisticated tasks. Finally, decoding might perform better on network-level measures rather than the LFP power features we used, based on recent studies linking connectivity metrics to cognitive control^{19,26–28}. At present, connectivity is difficult to estimate in real time or on a therapeutic device, although approaches are emerging⁴⁶.

Closely related, both this study and our earlier work operationalized cognitive control through the MSIT, which is only one of several tasks commonly used to measure control. This raises the question of whether our results could be generalized to other cognitive control metrics. Three factors suggest they could. First, there were substantial differences in task and stimulation details between this and our earlier study, but behavioural effects were consistent. Second, MSIT is primarily a combination of two other paradigms, the Simon

and Flanker tasks⁴⁷. Prior studies that separated these two conflict types showed consistent neural effects between them^{35,48}, suggesting that the same circuitry is involved regardless of the specific stimuli used to evoke conflict. Third, and closely related, a wide variety of cognitive conflict tasks consistently evoke theta-band frontal oscillations, implying a common mechanism of cognitive control across tasks^{19,20,24,28,43,48}. We observed and modulated those oscillations, suggesting that we engaged circuits that are related to cognitive control generally, not to our specific task. These are reasons for hope, but definitive demonstration that this approach applies to the broadest definitions of cognitive control will require testing across multiple tasks, settings and possibly species. There is an open question of whether modulating this task-related activity can improve cognitive control in more ‘real world’ situations, which will need to be tested as part of future clinical studies.

Translating these findings to practical clinical use will require overcoming both technical and ethical hurdles. Technically, there is substantial interest in the idea of cognitive control as a cross-cutting symptom of numerous psychiatric disorders^{19,39,49}, and thought leaders in psychiatry are urging the development of treatments that directly target decisional or cognitive dysfunctions^{50,51}. Another group recently reported closed-loop control of impulsivity, a close cousin of cognitive control⁵², and has advanced that approach into clinical trials⁵³. We and others have argued that cognitive remediation is a mechanism of multiple existing neurostimulation therapies, but is not recognized as such^{8,34,54}. However, there are no well-established diagnostic and monitoring scales for cognitive control or most other cognitive constructs. There are promising first steps^{39,55}, but establishing clinically valid assessments would be important. It would similarly be both important and interesting to understand how these results might change with a different task that assesses different aspects of cognitive control, for example, a more explicit extra-dimensional set shift³⁴. Given that control depends on a distributed frontostriatal network^{19,56}, there are probably multiple access points into that network. Broader explorations, including in animal models⁵², may clarify which stimulation targets are most useful in which situations.

Based on two participants reporting subjective well-being during our experiments, it seems possible that improved control might be directly reflected in standard mood or anxiety scales, but this would be a very noisy metric. Objective measures of cognitive control would also be important to ensure that the first use of such a technology rests on strong ethical grounds. Patients have a strong interest in closed-loop neurotechnologies⁵⁷, but there are broad societal concerns about altering personalities or authentic selves through such technologies^{58,59}. Given that we could improve task performance in patients with no overt cognitive impairment, these results also raise concern for attempts at cognitive enhancement in healthy humans. Given the numerous concerns surrounding potential enhancement⁶⁰, there should be careful ethical consultation before this approach is used outside of a rigorous and restricted research setting.

In summary, we have developed methods for real-time monitoring of human cognitive control-task performance, detection of lapses and closed-loop remediation of those lapses. We produced potentially useful cognitive and emotional effects with far less energy than conventional deep brain stimulation. The same framework could also be applied to other cognitive or emotional problems, for example, monitoring and enhancement of learning^{61,62} or emotion dysregulation^{63,64}. Although substantial technology gaps remain before these results can be directly applied in the clinic, and the evidence base needs to be built for cognition as a primary focus of treatment, our results could be the basis of a highly specific approach for intervention in human neuropsychiatric disease. In theory, and with further development, this approach could address a wider range of disorders than existing neurostimulation.

Methods

Experimental design. Twenty-one participants (age range: 19–57, mean age: 35, female: 12/21, left handed: 5/21) with long-standing pharmaco-resistant complex partial seizures were voluntarily enrolled after fully informed consent according to US National Institutes of Health and Army Human Research Protections Office guidelines. Consent was obtained by a member of the study staff who was not the participant's primary clinician. Study procedures were conducted while participants underwent inpatient intracranial monitoring for seizure localization at the Massachusetts General Hospital or the Brigham and Women's Hospital. The electrode implants were solely made on clinical grounds and not tailored for research purposes. Informed consent and all other ethical aspects of the study were approved and monitored by the Partners Institutional Review Board. Participants were informed that their involvement in the study would not alter their clinical treatment in any way, and that they could withdraw at any time without jeopardizing their clinical care. This was an exhaustive sample of all participants who were available, consented to research and had electrodes in brain regions of

interest. There was no pre-planned power calculation. Data collection ceased at the end of a continuous 4 yr period once study funding lapsed. Detailed clinical information on participants is in Supplementary Table 1.

The core hypothesis, that internal capsule stimulation would enhance cognitive control (shorten response times in a cognitive control task without altering error rates) was pre-specified based on our earlier work¹⁸. The analyses of behavioural and electrophysiological data, including the state-space and neural decoding modelling described below, were similarly pre-planned. The analyses described up through Fig. 2 were specified to replicate the earlier study and demonstrate that the effects of internal capsule stimulation were robust to changes in study population and stimulation details. Participants were part of a larger multi-modal study of the neural basis of mental illness¹⁹. They performed multiple other cognitive tasks both before and during their hospital stay and completed self-report scales related to emotional and cognitive difficulties. The list of tasks and scales and our strategies for comparing study participants against a normative database are described in ref. ¹³. All are well-validated assessments that have previously been used in large-scale studies. The specific comparison of these scales against participants' experience of subjective improvement was not pre-planned, as we did not know in advance that participants would report these effects.

Since we did not have a pre-specified stopping rule, we performed a post hoc sensitivity analysis in G*Power 3.1 (ref. ⁶⁵) to verify that we were adequately powered for the pre-specified hypothesis that capsular stimulation would improve cognitive control. The regression coefficients reported in Supplementary Table 2 and interpreted as significant in the text correspond to a partial R^2 of 0.0045 for the RT model and 0.01 for the theta-power model. These correspond to effect sizes f^2 of 0.0045 and 0.01, respectively. With the sample counts or degrees of freedom as in those tables, we have over 87% power for the detected effect sizes.

Behaviour paradigm – MSIT. Participants performed the MSIT⁴⁷ with simultaneous recordings of behaviour and LFPs from both cortical and subcortical brain structures. MSIT is a cognitive control task known to induce statistically robust subject-level effects at both the behavioural and neural level^{18,47,66}. These relatively large effect sizes amplified our ability to detect stimulation-induced differences by increasing task-related behavioural and neural signatures. MSIT trials consisted of three numbers between 0 and 3, two of which had the same value (Fig. 1a). Participants had to identify, via button press, the identity of the number that was unique, not its position. Each trial contained one of two levels of cognitive interference or conflict. Low conflict or congruent (C) trials had the unique number in a position corresponding to its keyboard button, and flanking stimuli were always '0', which is never a valid response. High conflict, or incongruent trials (I), had the position of the unique number different from the keyboard position, requiring execution of a non-intuitive visuo-motor mapping (Simon effect). On high-conflict trials, the non-unique numbers were valid responses (Flanker effect). To reduce the formation of response sets, we pseudo-randomized the trial sequence such that more than two trials in a row never shared the same interference level or correct response finger. This forces frequent strategy shifts and increases attention demands, which in turn increase the need to engage or deploy cognitive control. Each participant performed 1 to 3 sessions of MSIT. Each session consisted of multiple blocks of 32 or 64 trials, with brief rest periods in between blocks. These periods ranged from 0.9 to 57.6 min; the median break time was 5.25 min and 89.6% were under 10 min. During blocks, participants were instructed to keep their first through third fingers of their right hand on the response keys corresponding to the numbers 1–3. They were instructed to be as fast and as accurate as possible. Stimuli were presented for 1.75 s, with an inter-trial interval randomly jittered within 2–4 s. Stimuli were presented on a computer screen with either Presentation software (Neurobehavioral Systems) or Psychophysics Toolbox⁶⁷.

Electrophysiologic recording. We recorded LFPs from a montage of 8–18 bilaterally implanted depth electrodes (Fig. 1c, left and Extended Data Fig. 1). The decision to implant electrodes and the number, types and location of the implantations were all determined on clinical grounds by a team of caregivers independent of this study. Depth electrodes (Ad-tech Medical or PMT) had diameters of 0.8–1.0 mm and consisted of 8–16 platinum/iridium contacts, each 1–2.4 mm long. Electrodes were localized by using a volumetric image co-registration procedure. Using Freesurfer scripts (<http://surfer.nmr.mgh.harvard.edu>), the preoperative T1-weighted magnetic resonance imaging (MRI) scan (showing the brain anatomy) was aligned with a postoperative computed tomography (CT) scan (showing electrode locations). Electrode coordinates were manually determined from the CT scan⁶⁸. The electrodes were then mapped to standard cortical parcels or regions using an automatic, probabilistic labelling algorithm⁶⁹. Intracranial recordings were made using a recording system with a sampling rate of 2 kHz (Neural Signal Processor, Blackrock Microsystems). At the time of acquisition, depth recordings were referenced to an EEG electrode placed on the skin (either cervical vertebra 2 or Cz).

Open-loop capsule stimulation. We delivered electrical stimulation to either the dorsal or ventral internal capsule and surrounding striatal nuclei (Fig. 1b, right). We stimulated only one site in each block. We compared dorsal and ventral stimulation because this portion of the internal capsule has a well-described dorsoventral

topography^{29–31}. More ventral capsule fibres tend to originate in more ventral aspects of the PFC, particularly the orbitofrontal cortex. More dorsal fibres, in contrast, tend to originate in dorsolateral and dorsomedial PFC and in the dorsal cingulate. The latter structures are strongly implicated in cognitive control, particularly during conflict tasks^{19–21,38,56}. Thus, we hypothesized that dorsal stimulation would be more effective than ventral stimulation in this specific task. We did not have a pre-specified hypothesis regarding left versus right stimulation, but tested hemispheres separately because our past clinical studies of capsular stimulation suggested lateralized effects¹⁰. We did not test any other stimulation sites for their effects on MSIT performance because we had a specific hypothesis regarding the internal capsule. Other stereo-EEG (sEEG) electrodes were stimulated in separate experiments in these same participants, but those results are reported elsewhere^{63,71,72} and the outcomes of those experiments did not influence the present study.

We varied the order in which we stimulated the different sites (Supplementary Table 2). Stimulation experiments were performed towards the end of each participant's stay, when he/she was back on anti-epileptic medication, to reduce the risk of evoking a seizure. Such a session always started with 1 or 2 blocks of unstimulated trials and ended with an unstimulated block (Fig. 1d). In blocks with stimulation, it occurred on only 50% of the trials. The stimulated trials were chosen pseudo-randomly, with no more than three consecutive trials being stimulated. All participants received the same pseudo-random order of stimulated and unstimulated trials. The stimulation was a 600 ms long train of symmetric biphasic (charge balanced) 2–4 mA, 90 μ s square pulses at a frequency of 130 Hz. Stimulation was delivered through a neighbouring pair of contacts on a single depth electrode (bipolar), with the cathodal (negative) pulse given first on the more ventral contact. Stimulation was delivered by a Cerestim 96 (Blackrock Instruments), with parameters set manually by the experimenter and stimulation triggered by a separate PC that was either delivering or monitoring task or behavioural stimuli.

The stimulation frequency was chosen on the basis of a previous study¹⁸; it is also the frequency most commonly used in clinical DBS for psychiatric disorders. In contrast to that study, here we were able to harmonize stimulation parameters between participants because stimulation was not directly linked to medical treatment. All stimulation was delivered at the image onset to influence a decision-making process that begins with that onset. In all participants, before task-linked stimulation, we first tested stimulation at 1, 2 and 4 mA for 1 s repeated 5 times, with 5–10 s between each 130 Hz pulse train. We informed participants that stimulation was active and repeatedly asked them to describe any acute perceptions. We verified a lack of epileptiform after-discharges and ensured that the participants could not detect stimulation, for example, through unusual sensations. If participants reported any sensation (for example, tactile experiences in limbs or head), we limited task-linked stimulation to the next lowest intensity. In a secondary analysis, the recordings during these test stimulations were used to verify that dorsal and ventral stimulation sites had different patterns of cortical activation (see Extended Data Fig. 3).

Before and after all stimulation experiments, including closed-loop stimulation (see below), an experienced psychiatric interviewer (A.S.W.) asked participants to describe their current emotional state. We similarly prompted participants to describe any subjective experiences at arbitrary times throughout each stimulation experiment. All such reports were videotaped and transcribed. During the behavioural task runs, participants were fully blind to whether stimulation was active, that is, we did not inform them that we were beginning stimulated blocks.

Behaviour data analysis. The primary behaviour readout is the reaction time (RT), as all participants were over 95% accurate. First, we analysed the effects of stimulation and task factors at the trial level using a generalized linear mixed effect model (GLME), as in our earlier work¹⁸:

$$RT \sim \text{Conflict} + \text{blockStim} + \text{blockNum} + (1|\text{Participant})$$

This and all other GLMEs analysing RT data used a log-normal distribution and identity link function. Fixed effects in the GLME were Conflict (a binary variable coding the trial type as being low (0) or high (1) conflict), stimulation site (blockStim), and block number (blockNum) to account for fatigue or practice effects. Stimulation (blockStim) was coded at the block level, that is, whether the stimulation site in a given block was dorsal versus ventral capsule or left versus right, not whether stimulation was on versus off on a given trial. Block-level coding was a more parsimonious fit to the data (AIC = −449.3 for block-level coding, −359.7 for trial-level coding). Participant was a random effect. All categorical variables were automatically dummy-coded by MATLAB's 'fitglm' function. We excluded trials with missing responses and those with incorrect responses.

We further tested for a possible interaction between stimulation and the trial-to-trial conflict level by fitting an alternate model with an interaction term:

$$RT \sim \text{Conflict} + \text{blockStim} + \text{Conflict} * \text{blockStim} \\ + \text{blockNum} + (1|\text{Participant})$$

We assessed this model against our primary GLME by comparing AICs, which decrease in models that are more parsimonious fits to the observed data.

To develop closed-loop control and neural decoding strategies, we needed to convert this block-level analysis to a trial-by-trial estimate of participants' RT. Because the task rapidly switched back and forth between trial types, however, RT on any given trial was influenced by changes in conflict or interference in addition to (putative) stimulation effects and random variability. We sought to separately measure these processes and their change in response to capsular stimulation. To achieve this, we applied a state-space or latent-variable modelling framework^{3,74}. In this type of model, the variables of interest (here, the 'true' baseline RT and conflict-induced slowing, without influence from stochastic RT variations) are not directly observable. They are assumed to influence an observable variable (the actually observed RT) through a functional scaling, with additive noise. A further advantage is that this class of models is Markovian – the estimate of the unobserved variables ('states') on any given trial depends only on the currently observed RT and the estimate of the states on the preceding trial. This allows efficient, trial-by-trial computation that is well suited to real-time monitoring and control. For exactly this reason, a specific class of state-space model, the Kalman filter, has long been used for neural decoding in motor brain–computer interface applications^{36,37,75}.

Here we used the COMPASS toolbox⁷⁴ for MATLAB to fit a model that extracts a trial-level estimate of the RT independent of conflict effects. This model takes the form:

$$\log y_{RT,k} = x_{\text{base},k} + I_{\text{conflict},k} x_{\text{conflict},k} + \varepsilon_k \quad \varepsilon_k \sim N(0, \sigma_{\varepsilon}^2) \quad (1)$$

where $y_{RT,k}$ is the RT on trial k , and the x_k are latent, unobserved variables that we have previously termed 'cognitive states'. $x_{\text{base},k}$ represents the expected RT in the absence of conflict or other external influencing factors, whereas $x_{\text{conflict},k}$ represents the expected effect of conflict on the RT. $I_{\text{conflict},k}$ is an indicator variable, such that $x_{\text{conflict},k}$ only affects the expected RT on high-conflict trials. The observation noise, ε_k would capture other non-structured processes that influence the trial-to-trial RT. This noise is Gaussian with mean ε_k and variance σ_{ε}^2 . Note that this model follows the same distribution or link assumptions as the static GLME above, namely a log-normal distribution. One of the differences between this modelling approach and the classic Kalman filter is that RT is markedly non-Gaussian. It can only have positive values and is very skewed – RT distributions across a range of cognitive tasks have long right tails^{74,76,77}. The log-normal assumption corrects for this. The latent variables were modelled as:

$$x_{\text{base},k} = a_1 x_{\text{base},k-1} + v_{1,k} \quad v_{1,k} \sim N(0, \sigma_{1,v}^2) \quad (2.1)$$

$$x_{\text{conflict},k} = a_2 x_{\text{conflict},k-1} + v_{2,k} \quad v_{2,k} \sim N(0, \sigma_{2,v}^2) \quad (2.2)$$

where a_1 and a_2 define the decay of the state variables over time. $v_{1,k}$ and $v_{2,k}$ are mutually independent white noise processes with zero mean and variance $\sigma_{1,v}^2$ and $\sigma_{2,v}^2$, respectively. That is, we assumed that these two processes can vary entirely independently of one another (even though stimulation may influence both). Because both the state values on each trial and the parameters linking those values to the RT are unknown, this model has no closed-form solution. COMPASS estimates model parameters through an iterative expectation–maximization algorithm⁷⁴. We verified convergence of this algorithm on all participants by inspection of the model likelihood plot. We also tested models that included both accuracy and RT as observable outputs; these yielded no significant improvement over RT-only models (see main text and Supplementary Fig. 5).

x_{base} can be thought of as encoding more general, overarching aspects of cognitive control, such as effortful attentional focus on task stimuli, maintenance of goals in working memory and preparation to inhibit a pre-potent response on incongruent or high-conflict trials. x_{conflict} , in that framework, represents the cognitive load of actually deploying the response inhibition in response to conflict. This is sometimes framed as a 'reactive' model of cognitive control. We derived this two-variable model from both theory and earlier work. We previously observed¹⁸ (and confirmed in this work, see above) that capsular stimulation affects both high- and low-conflict trials. That is, it mainly alters the expected RT x_{base} . We considered, however, that reactive control in response to conflict might also be affected, and wanted to specifically test for that potential effect. Further, in our earlier work we found that this two-state model was necessary to accurately model RT during this same task⁷³. We verified that the two-state model was an appropriate fit to our data by comparing the RT residuals to those expected from a white-noise process. Unmodelled variance (for example, a third cognitive process not captured in our model) would lead to structured residuals that violate white-noise expectations.

The state-space model assumes that cognitive states are slowly varying, that is, they show a strong autocorrelation. (This was true in practice; the estimated values of a_1 and a_2 were close to 1 for all participants.) We thus cannot use the GLME to analyse stimulation-induced change in these latent variables (x_{base} , x_{conflict}) because they strongly violate the GLME assumption that individual data points are independent. We instead used non-parametric permutation testing, which is well-established as a method for inferential statistics on autocorrelated time-series⁷⁸. The stimulation labels of individual blocks were shuffled 1,000 times, with the shuffling nested within individual participants. This created a distribution

of cognitive state values under the assumption of no difference between stimulation sites (or between stimulated and non-stimulated trials). From that distribution, we inferred the P value of the actual state values under stimulation. For both the raw RT GLME and the cognitive state permutation tests, we compared up to four stimulation sites in each participant to baseline (no stimulation, NS1). Within each analysis, we corrected the P values for these multiple comparisons using an FDR step-down procedure via MATLAB's 'fdr' function.

Closed-loop stimulation. In three participants, we used that same state-space model to implement closed-loop control. First, for each participant we estimated model parameters by running the expectation–maximization fitting algorithm on 1–3 d of previous MSIT performance without brain stimulation. We used all available data for each participant to fit these models. These parameters were then provided to a real-time engine (also based on COMPASS, specifically the real-time decoding function 'compass_filtering') that estimated x_{base} and x_{conflict} on each trial. This specifically leverages the Markovian nature of our model in that estimating these states on each new trial is extremely fast once the model parameters are initially estimated.

We attempted to control x_{base} , which we considered to track the overall difficulty of sustaining attention and exerting cognitive control during conflict (more difficulty leading to longer RTs). In that framework, cognitive control enhancement would be reflected in a decrease in x_{base} . To achieve this, if the estimate on trial k was above a manually selected threshold, the system delivered electrical stimulation at the time of image or stimulus presentation on trial $k+1$. The threshold and stimulation effects were visible to the experimenter, but not to the participant, through a custom graphical interface running on a separate laptop. We set the threshold to attempt a decrease in x_{base} from its unstimulated value. The supervising experimenter (ASW) observed the participant's best performance from real-time state estimation during the initial non-stimulation block (64–128 NS1 trials). This process estimated both the mean (maximum likelihood value) of x_{base} , plus a confidence bound. We then set the threshold/target for x_{base} to be below that lower confidence bound, that is, to be outside the range of possible values that were consistent with the baseline or unstimulated performance. There was no quantitative rule for the distance between the estimated state bound and the control target; it was selected ad hoc to be outside our subjective estimate of overall process variability. Stimulation parameters and hardware were identical to the setup for open-loop stimulation. Because of limited experimental time, we could do either open-loop or closed-loop stimulation experiments with a particular participant, but not both. We attempted closed-loop control using the stimulation site that had the largest behavioural effect during open-loop experiments in the six previous participants. When that site was unavailable (not implanted in a given participant), we used the next best choice that was available.

For analysis of the closed-loop stimulation results, we re-ran the complete state-space estimation offline over the whole dataset, rather than using the less-accurate state values estimated in real time. A key difference is that the offline estimation contains a forward (filtering) and backward (smoothing) pass, allowing future data to influence each trial's estimate non-causally. By considering more information, this offline estimate more accurately reflects the 'true' cognitive process and its change in response to stimulation. To directly compare closed-loop and open-loop stimulation, we normalized the state values between these two runs such that the unstimulated blocks in both paradigms had a mean value of 1. That is, both open-loop and closed-loop results were expressed as change versus the unstimulated condition on the same day.

Neural data analysis – pre-processing. LFPs were analysed using a custom analysis code in MATLAB (Mathworks) based on FieldTrip (<http://www.fieldtriptoolbox.org>). To reduce the influence of volume conduction⁷⁹, LFPs were bipolar re-referenced by subtracting those recorded at consecutive electrode contacts on the same electrode shank. LFP was recorded from electrode pairs spanning 16 brain regions: prefrontal subdivisions, cingulate, orbitofrontal, temporal and insular cortices, amygdala, hippocampus, nucleus accumbens and caudate (Extended Data Fig. 1). All LFP data were decimated to 1,000 Hz and de-measured relative to the entire recording. Line noise (60 Hz) and its harmonics up to 200 Hz were removed by estimating noise signals through narrow bandpass filtering, then subtracting those filtered signals from the original raw signal. We removed pathological channels with interictal epileptiform discharges (IEDs). We detected such channels with a previously reported algorithm that adaptively models distributions of signal envelopes to discriminate IEDs from normal LFP⁸⁰. We then used a Morlet wavelet decomposition to estimate power in 6 frequency bands (4–8, 8–15, 15–30, 30–55, 65–110 and 135–200 Hz). This decomposition used a 1 Hz frequency resolution and 10 ms timestep, with the default parameters of 7 cycles per wavelet and a Morlet/Gaussian width of 3 standard deviations. We fractionated the high gamma (65–200 Hz) band into lower and upper bands to bypass the stimulation frequency at 130 Hz and a 60 Hz harmonic at 120 Hz. All time-frequency decomposition was performed after individual trials or epochs were cut from the continuous recording so that artifacts on stimulation trials could not influence power calculations on non-stimulation trials. Each trial was cut with a buffer of 2 s on each side to mitigate edge effects in the wavelet transform; data from these buffers were discarded before analysis.

Neural data analysis – mid-frontal theta power. Exercise of cognitive control is associated with higher theta (4–8 Hz) power in a frontocingulate network^{18,20,27,28,48}. We have specifically reported that stimulation in the internal capsule increases that task-evoked theta¹⁸. As an initial manipulation check, we sought to replicate these earlier results. We analysed an epoch of 0.1–1.4 s after image onset, which covered the decision-making period up to the median RT. We focused this analysis on non-phase-locked oscillations, which are more strongly correlated with cognitive conflict effects than are their phase-locked counterparts^{24,81}. From the target epoch, we subtracted the time-domain evoked response (ERP), which contains all phase-locked activity. We calculated this ERP separately for high- and low-conflict trials, and subtracted the appropriate ERP from each trial's time-domain data. We then transformed the time-domain to a time-frequency representation as above. We averaged power in our analysis epoch within the theta band. For visualization, we normalized this power as a log ratio relative to a baseline period of 0.5 s preceding image onset. For analysis, this log transformation was built into the Generalized Linear Model (see below).

To verify whether higher conflict evoked higher frontal theta, we analysed the blocks without stimulation. This avoids confounding effects of stimulation and conflict. For each participant, we pre-selected PFC channels that had a significant increase over baseline in task-evoked theta (t -test with threshold of $P < 0.05$ uncorrected). For this initial pre-screening step, to avoid a circular analysis, we did not split trials into high/low conflict. Rather, we identified channels that showed a general theta-band response to performing MSIT. In this reduced set of channels, we then divided the trials into low and high conflict, then computed the non-phase-locked theta power, as noted above. We combined all pre-selected channels in each PFC region, and for each region we fitted the GLME: Theta ~ Conflict + (1 | Participant), where Conflict is a binary variable coding the trial type as being low (0) or high (1) conflict. This and all other GLMEs analysing LFP power data used a log-normal distribution and identity link function. We FDR-corrected the resulting P values for testing of multiple PFC regions.

We then tested whether open-loop capsular stimulation caused a significant increase in theta in the unstimulated trials within a stimulation block (NS2) compared to those in the unstimulated blocks (NS1; Fig. 1c,d). To accurately assess stimulation effects, we discarded stimulation trials that we presumed were substantially contaminated by artifact. We then compared two types of non-stimulated trials (Fig. 1d). NS1 trials were from blocks in which no brain stimulation was given on any trial. NS2 trials were from blocks with stimulation, but were the pseudo-randomly selected 50% of trials that did not receive stimulation. These NS2 trials were artifact-free, but still showed the behavioural effect of stimulation, and thus should also show physiologic changes related to that behaviour change. We therefore tested whether the normalized theta power in NS2 trials was significantly greater than that in NS1, again using a GLME: Theta ~ blockStim + (1 | Participant).

For this model, we chose one PFC channel for each participant that had the highest theta during NS2 trials (regardless of conflict level or stimulation site, again to avoid circular analysis). P values were again FDR-corrected to control for testing of multiple stimulation sites against non-stimulation.

Neural data analysis – decoding. To effectively treat psychiatric disease, closed-loop stimulation will ultimately need to be applied outside of a structured task-based paradigm, that is, as patients go about their daily lives. That, in turn, requires the ability to detect lapses in cognitive control and conflict responses directly from brain activity. We thus developed a neural decoder for our cognitive state variables. We applied a neural encoding–decoding analysis with automatic feature selection, as in ref. ⁷³. The decoded variables were x_{base} and x_{conflict} from the model in equation (1). The neural features used for decoding were the LFP power in the above-mentioned frequency bands, averaged over a 2 s interval starting at the MSIT image onset. We broadened the analysis beyond the theta band because previous literature suggests that while theta is strongly associated with cognitive control, other frequencies also carry substantial information about task performance²⁶. The 2 s epoch was chosen to include both the response and post-response processing. This wider window produced smoother features with less trial-to-trial variance, improving decoder stability. Here we averaged only across a 2 s time interval (200 samples) to get power features per trial. Similar to the theta analysis, the encoder model fitting considered only NS1 and NS2 trials to prevent the influence of stimulation artifact. We focused on LFP spectral power (rather than other potential behavioural covariates, such as connectivity or coherence) because power can be efficiently computed within currently available implantable neural devices^{40,82,83}. Successful decoding of task performance from power alone could pave the way for use of these closed-loop controllers in clinical settings.

Decoding analyses were performed with out-of-sample validation using both stimulated and unstimulated MSIT datasets. For each participant's data, 20–66% of the total trials was used to fit an encoding model (training set). These consisted of NS1 trials in unstimulated datasets and both NS1 and NS2 trials in the stimulated experimental datasets. The training trials were selected from contiguous blocks of trials that, collectively, covered the full range of the states during an experiment. The encoding model that we used is a linear model of the form $Y_k \sim 1 + \beta x_k$, where Y_k is a neural feature and x_k is one of the cognitive states on the k -th trial. We

considered a feature to be a candidate for decoding if the modified F statistic⁷³ of the corresponding model corresponded to $P < 0.01$ (uncorrected). This procedure selected a set of candidate neural features that potentially independently encoded each cognitive state. The exact number of training trials for each dataset was determined as the minimum required to have a non-zero number of features selected by the encoding procedure.

Next, to reduce overfitting we pruned the selected set of neural features. The validation set for this pruning used 21–50% of the dataset for each participant, and had 25–50% overlap with the training set. We allowed this overlap to ensure that the validation set was sufficiently large for model pruning without leaving too few trials available for the subsequent test set (see below). On this validation set, we estimated the posterior distribution of the cognitive state from the neural data through a Bayesian filtering process similar to that used to estimate these same latent variables from RT⁷³. We calculated the root mean square error (RMSE) between the neurally decoded state and the ‘true’ (estimated from behaviour) cognitive state in this held-out test set⁷³. We then sequentially dropped the feature whose removal led to the most improvement in RMSE. The final decoder was then the set of features that survived this dropping step, that is, where dropping any further feature would increase RMSE on the validation set.

Finally, we reported the decoder’s performance on a fully held-out test set, namely the 18–46% of remaining trials from each participant that had not been used for either decoder training or feature pruning. The exact number of trials used for each participant’s decoder fitting, pruning and testing is given in Supplementary Table 6. A key challenge in performing these analyses was that the latent cognitive states (x_{base} and x_{conflict}) are themselves multivariate Gaussian estimates. The estimate’s value can depend on the starting point of the expectation–maximization process used to fit the state-space model. To control for this, we re-ran the behavioural estimation for each participant 1,000 times with different random seeds, producing 1,000 estimates of the underlying trajectory. (Further details are given in ref. ⁷³.) We then evaluated the neural decoder’s performance on the basis of whether its point estimate of the decoded state was within the confidence interval derived from these multiple trajectories.

We fitted this encoding–decoding model separately to data from unstimulated sessions (consisting of only NS1 trials), as well as to stimulated sessions (both NS1 and NS2 trials) to determine how the encoding structure was altered by electrical stimulation. We did not include stimulated trials in this analysis because there is a prominent stimulation artifact that makes these trials easily discriminable. In cases of stimulation–behaviour correlation, behaviour could be trivially decoded simply by detecting the artifact.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The main data supporting the results in this study are available within the paper and its Supplementary Information. Pre-processed and anonymized neural and behavioural data are available through Zenodo at <https://zenodo.org/record/5083120#.YOhvWehKiUk> and <https://zenodo.org/record/5085197#.YOhtouhKiUk>.

Code availability

Analysis code is available at <https://github.com/tne-lab/MSIT-Nature-Biomedical-Engineering>. The closed-loop neurostimulation system has been released as open-source code and documented⁴⁶, and the neural decoding and state-space modelling engines have similarly been released for open download (<https://github.com/TRANSFORM-DBS/Encoder-Decoder-Paper> and <https://github.com/Eden-Kramer-Lab/COMPASS>).

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Author contributions

A.S.W., D.D.D., E.N.E. and S.S.C. designed the study. I.B., A.Y., B.C., R.Z. and U.T.E. designed key software and tools required for data collection. K.K.E. and T.D. selected the psychometric scales administered to participants and provided unpublished data related to norming of those questionnaires. E.N.E. and G.R.C. performed all surgical procedures. A.S.W., I.B., B.C., R.Z., A.C.P., S.S.C. and D.S.W. collected data with participants during acute seizure monitoring. A.S.W., I.B., A.Y., A.C.P. and N.P. analysed data. I.B. and A.S.W. wrote the paper, with substantial inputs from A.Y., R.Z., A.C.P. and S.S.C. All authors had opportunities for critical input into and revision of the submitted manuscript, and approved its submission.

Competing interests

The authors declare no competing interests.

Additional information

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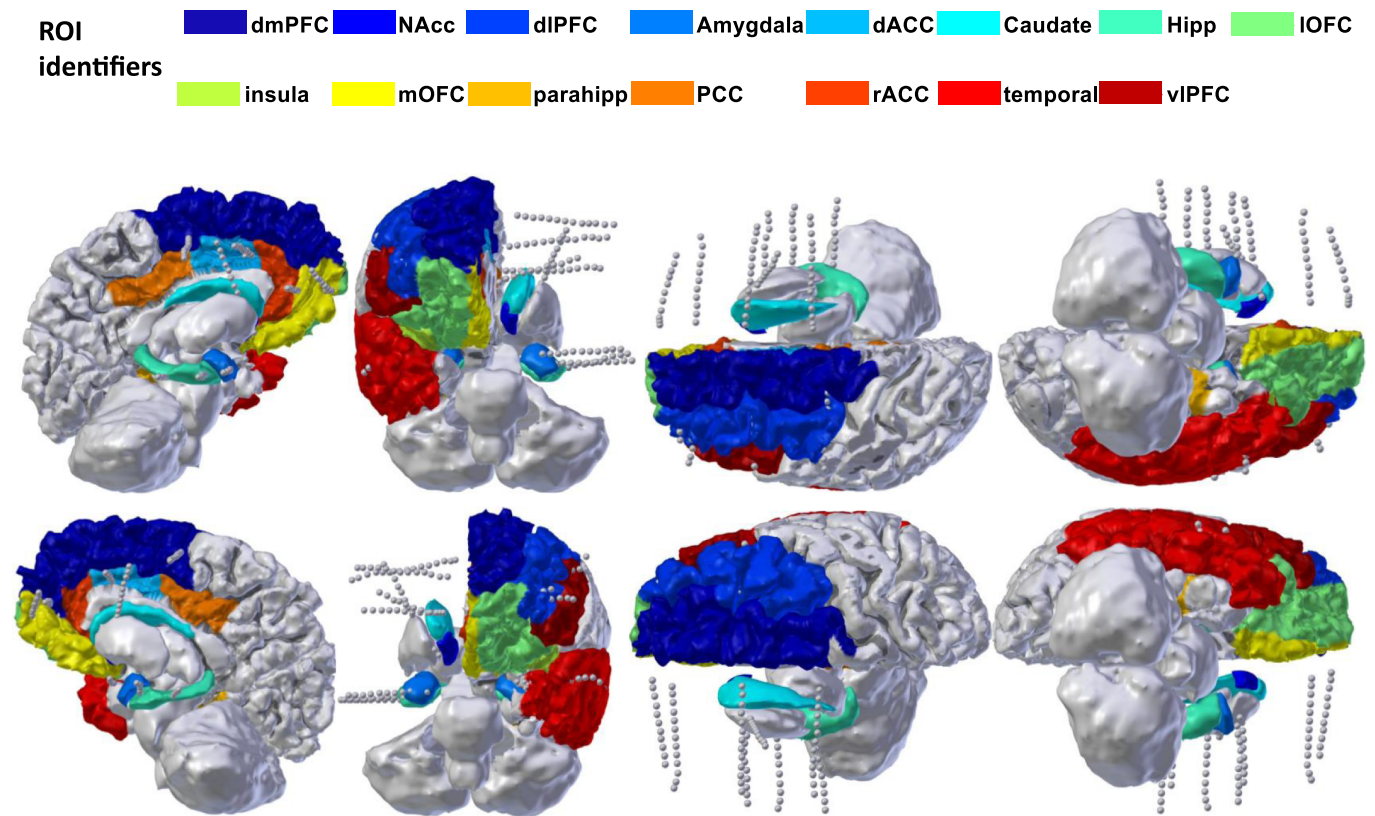
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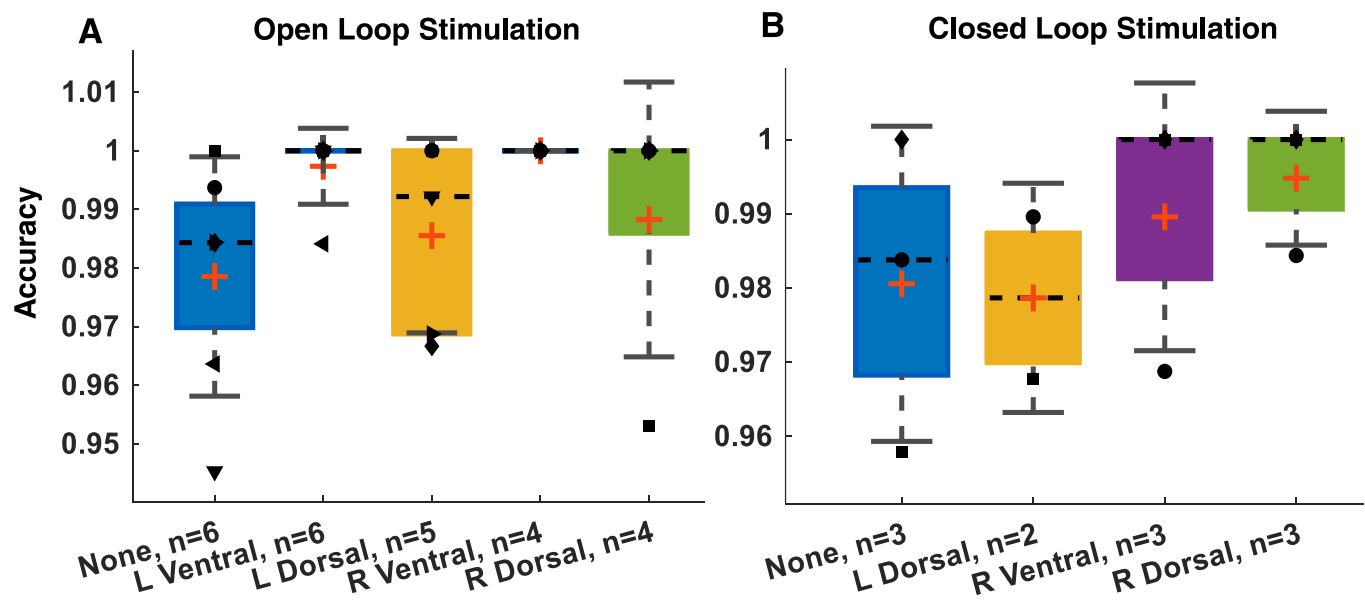
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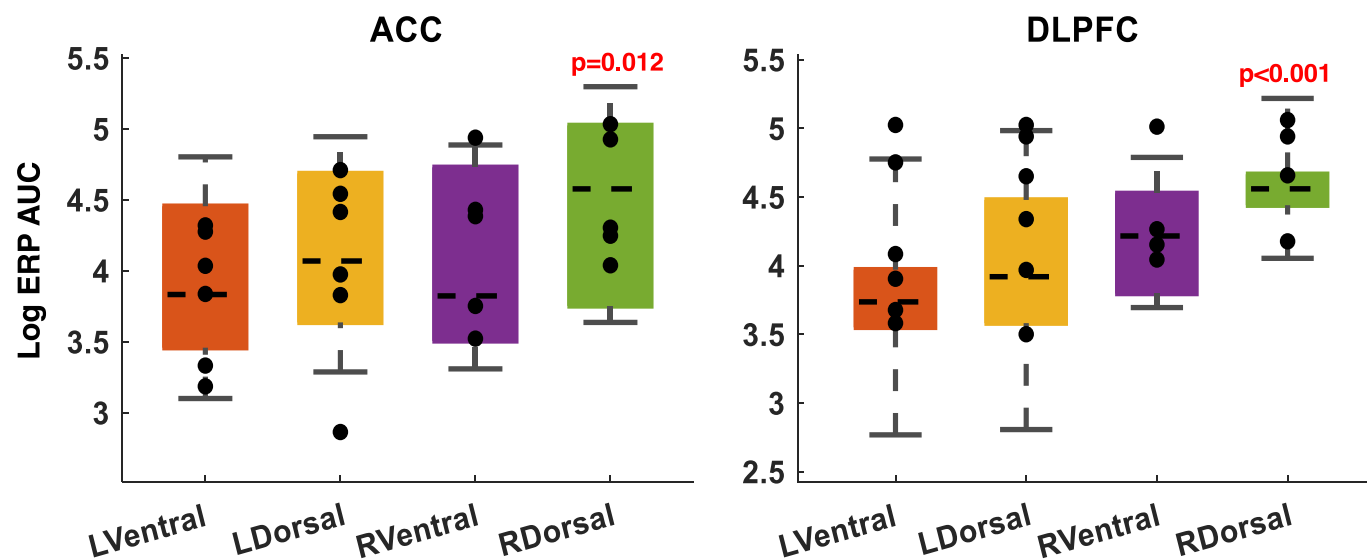
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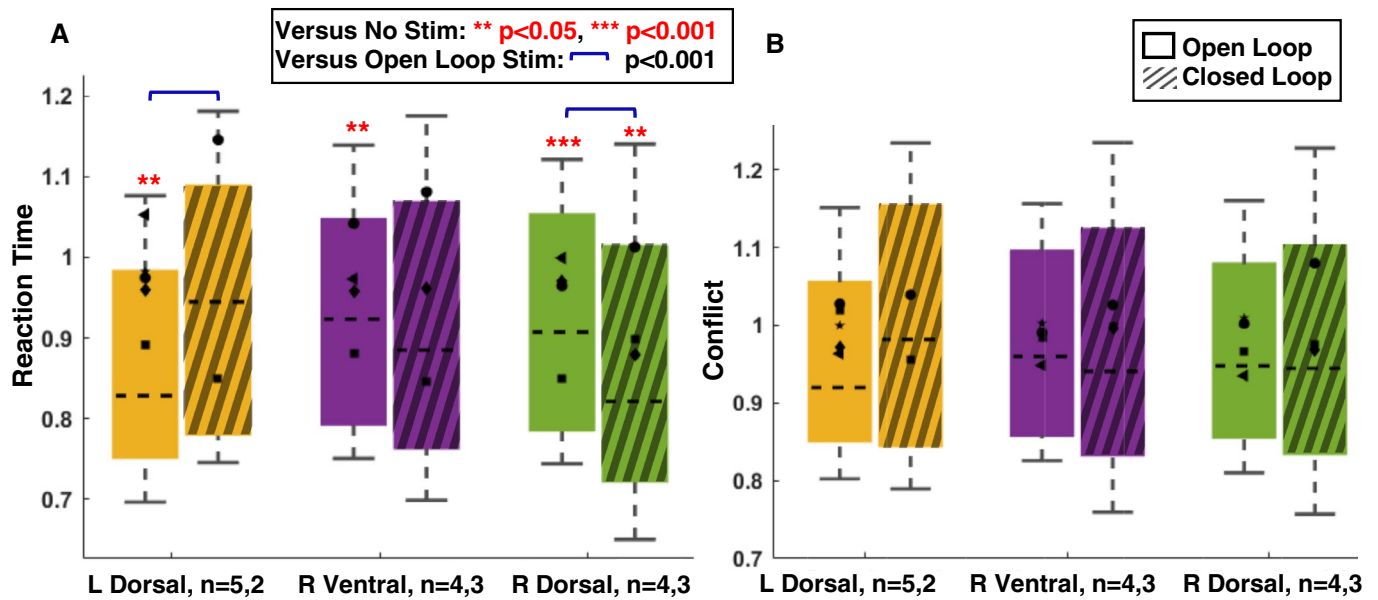
Extended Data Fig. 1 | iEEG Recording montage. Example recording montage from a single participant, with cortical parcellation overlaid. Electrode shanks represented by the grey dotted lines access a broad network covering multiple prefrontal structures, superficial and mesial temporal lobe, and striatum/internal capsule.



Extended Data Fig. 2 | Accuracy results. Accuracy during different stimulation experiments, for A) open-loop and B) closed-loop capsular stimulation. Boxes show the mean and confidence intervals for accuracy with stimulation at each site. Colours indicate stimulation sites as in the main text. The p-value above each bar represents a binomial exact test of accuracy compared to the non-stimulated baseline condition, with Benjamini-Hochberg false discovery rate correction. All accuracies are above 95%, with accuracy during stimulated blocks being very slightly higher in most cases. No results exceed chance significance. We did not have open- and closed-loop data from the same participants. To compare the CL and OL conditions, we therefore compared their accuracies across participants with a Fisher exact test for each of the three stimulation sites (L Dorsal, R Ventral, R Dorsal) that were used in both conditions. L Dorsal: $p = 0.645$. R Ventral: $p = 0.440$. R Dorsal: $p = 0.655$. These provide no evidence for a difference between OL and CL conditions. These results do not support a change in accuracy with any stimulation type. That is, the observed decrease in reaction times is a true performance improvement, not a shift along a speed-accuracy tradeoff. We were unable to analyse accuracy in the GLME framework because the differences between stimulation sites are so small as to make the models non-identifiable in all cases.



Extended Data Fig. 3 | Cortical response to internal capsule stimulation. Topographic structure of the internal capsule yields differential cortical effects from stimulation at different capsular sites. Before task-linked stimulation, we performed safety/perceptibility testing, where we repeatedly stimulated each potential site with brief 130 Hz pulse trains (see Methods). Each of those trains created an evoked response potential (ERP) in various cortical regions. For each participant, we collected all sEEG channels that were localized to grey matter of DLPFC or ACC. We then quantified the post-train ERP as the sum of the area under its polyphasic curve (AUC). We limited this analysis to channels ipsilateral to the site of stimulation. Each marker represents the mean log(AUC) in one participant. Boxes show the mean and confidence intervals for the ERP AUC from stimulation at each site. The stimulation sites that were more effective behaviorally produced the largest ERPs in these cognitive-control-associated regions, with right dorsal stimulation having the largest effects. (p-values represent t-test on the regression coefficients of a log-normal GLM, that is the same analysis used in main text Fig. 2). In the left hemisphere, dorsal stimulation produced larger responses than ventral stimulation, but this did not reach statistical significance given the small number of trials (5 test trains per participant). These results are consistent with the known topography of the internal capsule, where fibers that connect DLPFC and ACC to thalamus run in the dorsal-most part of the anterior limb, that is in close proximity to our chosen dorsal electrodes.



Extended Data Fig. 4 | Open-loop and closed-loop effects in manifest data. Effect of open-loop and closed-loop capsular stimulation on A) reaction time (RT) and B) Conflict related RT. Conflict related RT is calculated as the residual reaction time after subtracting the mean reaction time of the congruent trials in the same block, that is it has an expected value of 0 ms on non-conflict trials. We consider it as the closest raw/manifest data analogue of x_{conflict} . We note, however, that both of these manifest RT variables include the Gaussian noise that is removed by the state-space filtering that produces x_{base} and x_{conflict} . As such, the data in this figure are by definition noisier, and the analysis has lower statistical power. This leads to smaller effects in the open-loop results compared to main text Fig. 3. Closed-loop stimulation of the right dorsal internal capsule (our most effective open-loop intervention) was more effective than its open loop counterpart at reducing raw RT (the counterpart of x_{base}). Consistent with the specificity illustrated in main text Fig. 4 C, there was no advantage for closed-loop stimulation on the conflict-specific RT (the counterpart of x_{conflict}). All formatting and graphical elements follow the conventions of main text Fig. 4.

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Custom scripts, primarily MATLAB-based, some C++ and Simulink. All are available via GitHub (links under 'Data analysis' below).

Data analysis

Analysis code is available at <https://github.com/tne-lab/MSIT-Nature-Biomedical-Engineering>. The closed-loop neurostimulation system has been released as open-source code and documented, and the neural decoding and state-space modeling engines have similarly been released for open download (<https://github.com/TRANSFORM-DBS/Encoder-Decoder-Paper> and <https://github.com/Eden-Kramer-Lab/COMPASS>).

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Life sciences study design

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Sample size	We performed exhaustive sampling for all available participants during the study period. We carried out post-hoc power calculations to verify we had adequate power for the key reported effects.
Data exclusions	Neural data were excluded from channels or trials contaminated by artefacts. These were determined by pre-established rules described in detail in Methods.
Replication	This experiment is a replication of a prior report of similar effects in a different human-subjects population (but now adds the capability for closed-loop control). We believe this verifies reproducibility.
Randomization	Randomization was not relevant, as we employed a within-subjects design.
Blinding	Blinding was not relevant, as we employed a within-subjects design. The investigators working directly with the participant could not be blinded because the neurostimulation is obvious on the recording traces to anyone with training, and because all researchers needed to know the experimental condition to maintain participant safety in case of emergency.

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Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Human research participants

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Population characteristics	Twenty-one participants (age range: 19–57; mean age: 35; female: 12/21; left-handed: 5/21) with long-standing pharmacoresistant complex partial seizures.
Recruitment	All participants who underwent neurosurgery for epilepsy during the study period were approached. There were no known systematic sources of bias.
Ethics oversight	Informed consent and all other ethical aspects of the study were approved and monitored by the Partners Institutional Review Board.

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